

Modelling the thermal properties of cement by means of machine learning force fields

A MSci Project Report

by

Niall McCallion
40360191



School of Mathematics and Physics
Queen's University, Belfast

18th April 2026

Abstract

Understanding the thermal behaviour of cementitious materials is essential for improving the efficiency, durability, and sustainability of modern infrastructure. At the atomic scale, these properties are governed by the structure and dynamics of calcium silicate hydrate (C–S–H), the primary binding phase in cement. However, C–S–H is structurally disordered, making it difficult to model directly. Tobermorite, a naturally occurring crystalline analogue, provides a tractable model system, yet its thermal properties—most notably the Debye temperature—remain unreported in the literature. This project develops a Committee Neural Network Potential (C-NNP) for tobermorite-9Å, trained on density functional theory (DFT) and ab initio molecular dynamics (AIMD) data, to enable accurate and uncertainty-aware molecular dynamics simulations at near-DFT accuracy. The committee framework provides intrinsic uncertainty quantification and supports active-learning refinement of the potential energy surface. Using the resulting machine-learning force field, vibrational spectra, heat capacities, and the Debye temperature are extracted from long-timescale simulations. This work presents the first computational determination of the Debye temperature of tobermorite and demonstrates the capability of C-NNPs to model the thermal properties of complex cementitious materials, providing a foundation for future studies of energy-relevant behaviour in C–S–H and related phases.

Keywords: Tobermorite; Calcium Silicate Hydrate (C–S–H); Machine Learning Force Fields; Committee Neural Network Potentials; Density Functional Theory; Ab Initio Molecular Dynamics; Thermal Properties; Vibrational Density of States; Heat Capacity; Debye Temperature.

Acknowledgements

I would like to thank my advisors, Dr Lorenzo Stella and Dr David Wilkins. They made cement interesting, and ensured the completion of this project by pushing me every step of the way even when that meant reading emails that were sent at ungodly hours. They provided me a new perspective on cement and a new passion for AWK coding and I can not thank them enough for making this such an enjoyable research project. I could not have undertaken this degree without my family; especially my parents, who have always supported my academic journey and who have taught me to "never give up" and to always try my best and I hope this report shows them this. I also had the pleasure of working with my girlfriend Alanna Trimble, who has been by my side through the ups and downs of this project (literally next to me doing her own thesis). She has been a light in the dimly lit office, and together we have kept the other one going. The friends who have taken on the mathematics degree beside me, fighting through late-night libraries, have been an absolute pleasure over the 4 years and I could not have asked for a better group to do it with; including: Amy Watt (code breaker), Eva Loughridge, Cammeron Hedderman BSc, Erin Colgan BSc, Heather Archer, Amy Greer BSc, Sarah Halligan BSc and of course Oliver Boyce BSc. I would be remiss if I was not to thank my dear friend Nia Mallon who without computer science would have been a lonely and gruelling endeavour. Finally, I would like a special thanks to my housemates, their encouragement and willingness to listen to the chemistry of cement should be rewarded without (Aoife Leonard BSc, Beth Noble BSc pending, Emily Toolan BSc, Findlay Murray BSc and Lauren McShane MSci pending) having been there when I've returned from the library late at night and providing cups of tea, a movie and a chat. They have ensure that I gave myself time off and reminded me that I can complete this degree, without them this year would not have been as enjoyable as it has been.

Contents

Abstract	ii
Acknowledgements	iii
1 Introduction	1
1.1 Motivation	1
1.2 Literature Review	3
1.3 Project Aims and Scope	5
2 Theory	6
2.1 Molecular Dynamics	6
2.2 Density Functional Theory	8
2.2.1 Plane-Wave Basis and cut-off values	9
2.3 Neural Networks	9
2.3.1 Behler-Parrinello Method	10
2.3.1.1 Radial symmetry functions	11
2.3.1.2 Angular symmetry functions	11
2.3.2 Committee Neural Network Potentials	12
2.4 The δ -Learning Approach	13
2.5 Thermal Properties	15
2.5.1 Velocity Autocorrelation Function	15
2.5.2 Vibrational Density of States (VDOS)	16
2.5.3 Heat Capacity	16
2.5.4 Debye Model and Debye Temperature	17
2.5.4.1 Determination of the Debye Temperature	18
3 Methods and Development	19
3.1 Ab initio Calculations	19
3.1.1 Convergence study	19
3.1.2 AIMD	21
3.2 Generating NN models	23

4	Results and discussion	25
4.1	Re-training and evolving the committee	25
4.2	Extracting Thermal Properties	27
4.2.1	From VACF to Vibrational Density of States	28
4.2.2	Connection to Thermodynamic Properties	30
5	Conclusion	33
	Appendices	35
A	Extended XYZ-co-ordinates	36
B	Behler-Parrinello Symmetry functions	38
C	Justification and discussion of model development	40

Chapter 1

Introduction

1.1 Motivation

Cement is a complex material, one of the main components along with gravel or sand of concrete. Cement-based materials are among the most widely used manufactured materials in the world, forming the structural backbone of modern infrastructure for buildings, reservoirs, dams and nuclear disposal sites. Concrete however, contributes significantly to global energy loss. Buildings alone account for “somewhere between 20 and 40% of the globally generated energy and energy-related environmental emissions”, with a large contribution in regards to heating, ventilation and air conditioning systems [1]. Improving the energy efficiency of buildings can be achieved by designing new “smart” materials to integrate within cement [2]. Producing more durable and sustainable cement-based materials is therefore a key challenge in reducing carbon emissions and enhancing the sustainability of the built environment. Understanding cement, on an atomic level is crucial to help in the design of better cementitious materials.

Within these cementitious materials, the main component we wish to look at is the binding agent, calcium silicate hydrate (C-S-H), the nanostructured gel responsible for the strength, durability, and long-term behaviour of concrete. Where $C = CaO$, $S = SiO_2$ and $H = H_2O$ [3]. However, C-S-H is structurally disordered and difficult to model directly at the atomic scale. For this reason, computational studies frequently rely on naturally occurring minerals such as jennite or the tobermorite family ($9\text{\AA}$ $Ca_5Si_6O_{16}(OH)_2$ shown in Fig.(1.1), $11\text{\AA}$ $Ca_4Si_6O_{15}(OH)_2 \cdot 5H_2O$, and $14\text{\AA}$ $Ca_5Si_6O_{16}(OH)_2 \cdot 7H_2O$ where 9, 11, and 14 represent the distance of space between layers caused by different hydration degrees. As they share many features with C-S-H (Ca-O sheets bonded to silicate chains) they can be used as analogue crystals that are more computationally friendly due to controlled unit cells and well-defined geometries.

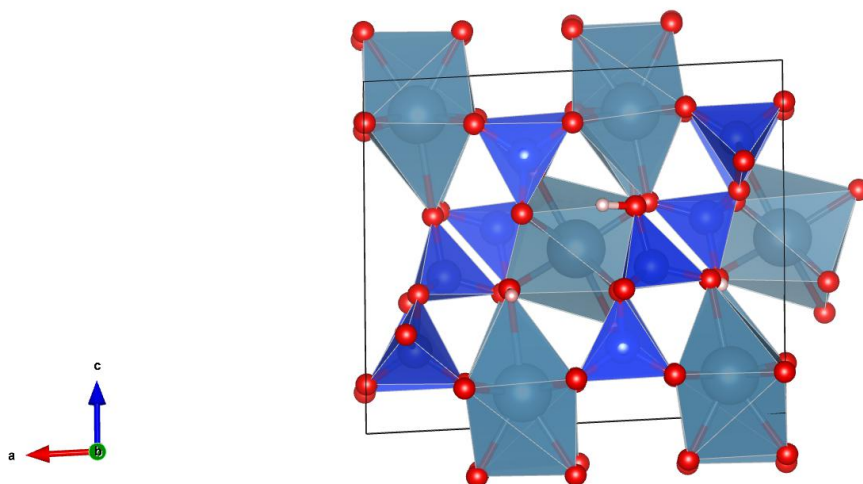


Figure 1.1: Tobermorite-9Å unit cell

The blue and light blue blocks in Fig.1.1, represent the SiO_2 tetrahedrons and CaO polyhedrons respectively, with the water molecules found in the interlayers. This data was taken from [2] and this structure will be used for the simulations within this project. However, modelling thermal properties accurately remains computationally challenging, as these properties require atomistic simulations over sufficiently long timescales to sample the relevant atomic motions. Molecular dynamics (MD) provides a suitable framework for such simulations, but its accuracy depends directly on the underlying potential energy surface (PES). In ab initio molecular dynamics (AIMD), the interatomic forces are evaluated at each time-step using density functional theory (DFT), which provides near-quantum-mechanical accuracy through an explicit quantum treatment of the electronic structure [4]. In this sense, AIMD is a quantum method, in contrast to classical MD based on empirical force fields. However, the nuclear motion in AIMD is typically treated classically, and quantum nuclear effects such as zero-point motion are neglected, an approximation that is generally well-justified at room temperature. Despite its accuracy, AIMD remains computationally expensive and is therefore limited to short simulations and small system sizes (order of 1000 in femtoseconds), making it unsuitable for extended thermal sampling in complex materials such as tobermorite.

This motivates the use of Machine Learning Force Fields (MLFFs), which learn the potential energy surface directly from DFT data. Neural Network Potentials (NNPs), particularly those based on the Behler–Parrinello framework [5], have shown remarkable success in modelling complex materials with near-DFT accuracy at a fraction of the computational cost. Prior work on tobermorite has demonstrated that NNPs trained on both energies and forces can reproduce DFT vibrational spectra with high fidelity [6]. Yet single-network NNPs suffer from a critical limitation: they provide no measure of uncertainty. When the model encounters configurations outside its training distribution, it may extrapolate unpredictably, leading to unstable MD trajectories or inaccurate thermodynamic predictions. This is especially problematic for materials like tobermorite, where subtle structural distortions and hydration-dependent behaviour require robust generali-

sation. This is where Committees of Neural Networks (C-NNPs) can be developed with papers showing C-NNPs improve accuracy, provide generalization error estimates, and enable active learning strategies within models [7].

1.2 Literature Review

The benefits of using MLFFs rather than the standard force fields has been long understood within literature:

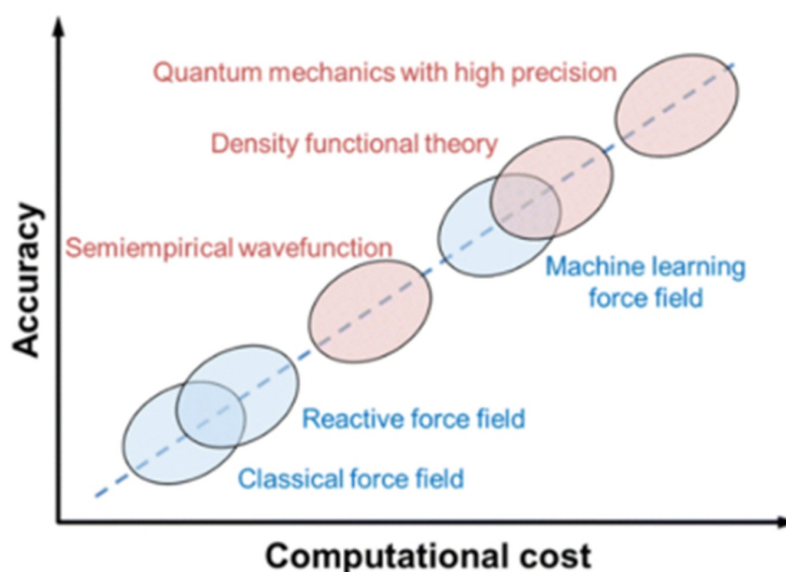


Figure 1.2: Comparison of the accuracy and computational cost of various quantum mechanics and force field methods [4].

Machine-learning interatomic potentials have become an increasingly important tool in materials modelling. Among them, the Behler–Parrinello neural-network potential (BP-NNP) is a widely adopted framework [5]. The BP approach constructs the total energy as a sum of atomic contributions, each determined by a feed-forward neural network [8], acting on a set of symmetry functions designed to encode the local atomic environment. These functions preserve translational, rotational, and permutational invariances, ensuring that the representation respects the intrinsic symmetries of the PES. Studies employing BP-NNPs have demonstrated their ability to achieve near-DFT accuracy in predicting structural, vibrational, and thermodynamic properties across a wide range of materials, including oxides and silicates [6].

Within Kobayashi et.al [6], The paper explores the topic of MD simulations enhanced by MLFFs using the BP method. In the paper 2 artificial NN’s were created on tobermorite minerals; one for both forces and energies (MLP-FE) and another optimized only on energies (MLP-E). The comparison of the 2 NN’s showed that the MLP closely reproduced properties (radial distribution and vibrational density of states), derived from DFT when using MLP-FE, whereas MLP-E showed significant deviations. The paper shows the scalability of the MLP-FE model to large scale MD simulations. These findings showed the importance of including interacting forces when training a NN, so for this reason a

neural network potential (NNP) based on Forces and energies will be developed in this study.

Despite their success, single NNP possess a key limitation: they do not provide an intrinsic means to quantify uncertainty. When these models encounter configurations outside their training distribution, they may extrapolate unpredictably, leading to unphysical force predictions or unstable MD trajectories [7]. This limitation is particularly significant in complex materials such as tobermorite, where subtle distortions in silicate chains or the vibrations within Hydrogen bonding can significantly alter local energetics.

A paper titled "Committee neural network potentials control generalization errors and enable active learning" [7] discusses the uses of C-NNPs which talks about how a committee addresses this issue by training multiple independent neural networks with identical architectures but different random initialisations, sharing the same atomic environment. Because each network converges to a different local optimum, the variance among their predictions serves as a measure of uncertainty. The measure of uncertainty of the committee can be calculated by measuring the standard deviation of the predictions of each individual members. This new C-NNPs outperforms the individual NNPs contained within it. The benefits of having a generalized error within the system can ensure the data is not over-fitted by a single model and is instead the best model across the full committee. As well as being a useful disagreement tool, committees can identify data within the simulations that can be used within active-learning workflows. An active learning approach is where high-uncertainty configurations encountered during MD are re-evaluated using DFT and added to the training set, further exploring the configuration space and improving the coverage of the PES and enhancing model stability.

The creation of a robust low error C-NNPs is crucial to the development of reliable chemical and structural properties. This can be a tedious task that can be improved by the use of δ -learning [9]. In this approach, the total energy is decomposed into a cheap computational baseline contribution, such as that from a Lennard-Jones potential, and a machine-learned correction. The neural network is trained only on the residual between the baseline and the DFT reference. This reduces the complexity of the function that the network must learn because less effort is spent collecting data and fitting configurations where only two atoms in a large structure are close to each other [9]. This leads to improved stability during MD, even when the Root Mean Square Error (RMSE) on test data is not minimised. δ -learning is particularly valuable in systems where the baseline potential already captures essential qualitative features of the PES, allowing the neural network to focus on the more subtle components.

The thermal behaviour of silicate minerals has been widely studied due to its importance for modelling vibrational properties, phase stability, and high-temperature transformations.

Krupka's comprehensive USGS report [10] outlines two principal approaches for determining Debye Temperature (Θ_D): low-temperature calorimetry and elastic-constant analysis. At sufficiently low temperatures, only acoustic phonons contribute to the heat capacity, leading to the characteristic Debye T^3 behaviour. Where at low temperatures the heat capacity is approximately T^3 . This method has been successfully applied to silicate minerals such as enstatite, wollastonite, and phenakite.

A second approach derives Θ_D from the mean sound velocity obtained from the elastic stiffness constants using numerical integration of the Christoffel equation.[10] For the purposes of this report low-temperature fitting will be used when calculating the Θ_D .

When looking at other C-S-H phases, which are structurally related to tobermorite.

Rodriguez et al [11]. provide a detailed investigation of synthetic C–S–H. Their results show that C–S–H undergoes a characteristic sequence of transformations transforming into wollastonite around $1000K$. with the transformation temperature increasing with C/S ratio. This behaviour is directly relevant to tobermorite, which also dehydrates and collapses into wollastonite at elevated temperatures [12].

Although Debye temperatures are well documented for many anhydrous silicates, no such values exist for tobermorite which may help further understand the structure and its properties for future work. Recent research has expanded the relevance of tobermorite beyond structural modelling. A 2023 study by Orisakwe et al [2]. Demonstrated that tobermorite exhibits promising thermoelectric behaviour, reporting Seebeck coefficients in the range of “200–600 V/K, indicative of a good thermoelectric material”. Their work shows that tobermorite phases behave as intrinsic p-type semiconductors due to the dominance of Si–O tetrahedral states near the valence band maximum. This suggests that cementitious phases may be engineered not only for structural performance but also for energy harvesting, reinforcing the importance of understanding their thermal and vibrational behaviour at the atomic scale.

1.3 Project Aims and Scope

To address this, the present project develops a C-NNPs potential for tobermorite-9Å. A committee model trains multiple independent NNPs on the same dataset and aggregates their predictions. The spread in their outputs provides a direct estimate of epistemic uncertainty, enabling: more robust force predictions, detection of poorly sampled regions of configuration space, active learning strategies to iteratively improve the model, and greater confidence in downstream thermal property calculations. By combining DFT, ab-initio molecular dynamics, Behler–Parrinello symmetry functions, and committee-based uncertainty quantification, this project aims to build a reliable, scalable, and uncertainty-aware MLFF for tobermorite-9Å. The model will then be used for simulations of vibrational spectra and heat capacity (From which the Debye Temperature can be calculated), properties essential for understanding the thermal performance of cement-based materials. By integrating first-principles calculations, advanced MD techniques, and machine-learning methods, this project aims to produce a reliable and uncertainty-aware MLFF for tobermorite-9Å. Validate its use in predicting thermal properties, which includes aiming to compute the Debye temperature of tobermorite using machine-learning force fields. Such a model supports extended MD simulations and provides a foundation for future investigation of thermal properties in cementitious materials such as composite compounds as seen in [2].

Chapter 2

Theory

Understanding the thermal properties of tobermorite-9Å requires a combination of first-principles electronic-structure methods, atomistic simulation techniques, and machine-learning approaches for constructing accurate interatomic potentials. This chapter outlines the theoretical foundations of the methods used in this work, beginning with molecular dynamics, followed by density functional theory, moving into the Behler–Parrinello machine-learning framework, committee neural networks, and δ -learning. Concluding with a discussion of various thermal properties that can be extracted from the model’s data.

2.1 Molecular Dynamics

Molecular Dynamics (MD) is a computational technique that is used to simulate the time evolution of a system of interacting atoms by integrating Newton’s equations of motion. It provides a link between microscopic atomic interactions and macroscopic thermodynamic behaviour, making it a fundamental tool for studying the vibrational, structural, and thermal properties of materials [13]. MD is essential in this project because the thermal properties of tobermorite, such as heat capacity, are derived from atomic trajectories evolving on a Potential Energy Surface (PES). Although classical MD relies on empirical potentials or force fields [14]. This work ultimately employs machine-learning force fields trained on *ab initio* data to achieve near-DFT accuracy at significantly reduced computational cost.

In classical MD, atoms are treated as point masses obeying Newton’s second law;

$$m_i \frac{d^2 \mathbf{r}_i}{dt^2} = \mathbf{F}_i, \quad (2.1)$$

where m_i and $\mathbf{r}_i$ denote the mass and position of the atom i , and $\mathbf{F}_i$ is the force acting on it. The forces are obtained as the negative gradient of the potential energy V ;

$$\mathbf{F}_i = -\nabla_{\mathbf{r}_i} V(\mathbf{r}_1, \dots, \mathbf{r}_N). \quad (2.2)$$

The potential energy surface (PES) V determines how atoms interact. Traditional force

fields are often too rigid due to their simplicity and are inadequate for modelling changes in atom connectivity such as bond breaking and formation during reactions, which also lead to a reduction in calculation accuracy [4]. Hence, why PES are used within this project, and how this is learnt is mentioned later.

Because analytical solutions to Newton’s equations are impossible for many-body systems, MD relies on numerical integration. The velocity Verlet algorithm is widely used due to its stability, time-reversibility, and good energy conservation [13]. To obtain NVT dynamics (constant particle number (N), volume (V), and temperature (T)), a thermostat must be coupled to the Velocity Verlet scheme. However the core of the Verlet equations remains largely the same. The algorithm updates position and velocities according to

$$\mathbf{r}(t + \Delta t) = \mathbf{r}(t) + \mathbf{v}(t)\Delta t + \frac{1}{2}\mathbf{a}(t)\Delta t^2, \quad (2.3)$$

$$\mathbf{v}(t + \Delta t) = \mathbf{v}(t) + \frac{1}{2}[\mathbf{a}(t) + \mathbf{a}(t + \Delta t)]\Delta t. \quad (2.4)$$

The time step Δt must be small enough to resolve the fastest vibrational modes. For materials such as Tobermorite, O–H bonds are the fastest mode and require time steps to be < 1.0 fs, to ensure that hydrogen dynamics are captured accurately [15], consistent with the time step used in this project ($\Delta t = 0.25$ fs).

MD simulations can be performed under different statistical ensembles depending on which thermodynamic variables are held constant. This project uses the canonical (NVT) ensemble, which is appropriate for equilibrating the tobermorite structure at a target temperature (e.g. 300 K) before generating training data.

To maintain a constant temperature, MD simulations employ thermostats that modify atomic velocities. This is achieved using the Canonical Sampling through Velocity Rescaling (CSVR) thermostat [16], which rescales atomic velocities using a stochastic term designed to ensure correct sampling of the canonical distribution.

To model bulk materials, MD simulations use periodic boundary conditions (PBCs). The simulation cell is replicated infinitely in all directions:

$$\mathbf{r}_i^{(n)} = \mathbf{r}_i + n_1\mathbf{a} + n_2\mathbf{b} + n_3\mathbf{c}, \quad (2.5)$$

where $\mathbf{a}, \mathbf{b}, \mathbf{c}$ are lattice vectors and $n_i \in \mathbb{Z}$.

These replication through configuration space eliminates surface effects and approximates as an infinite solid. For crystalline materials such as Tobermorite-9Å, this is essential to ensure physically meaningful behaviour.

Two main types of MD are relevant to this project; Classical MD uses empirical potentials to compute forces and is computationally efficient, enabling simulations over a longer period of time. As mentioned above, it fails to capture the complexity of the molecule. In contrast, *ab initio* molecular dynamics (AIMD) computes forces directly from Density Functional Theory (DFT) at each time step, providing quantum-mechanical precision at the cost of significantly higher computational expense [4]. AIMD is used in this work to generate high-quality reference data for training machine-learning force fields. Once trained, the machine-learning force field enables long timescale MD simulations with near-DFT accuracy.

2.2 Density Functional Theory

Density Functional Theory (DFT) is a quantum-mechanical framework used to determine the electronic structure of many-electron systems. It provides a practical alternative to solving the full many-body Schrödinger equation, whose computational cost scales exponentially with the number of electrons [17]. DFT forms the foundation of the *ab initio* calculations used in this project, supplying the reference energies and forces required to train the machine-learning potential.

The behaviour of a system of N_e interacting electrons is governed by the time-independent Schrödinger equation,

$$\hat{H}\Psi(\mathbf{r}_1, \dots, \mathbf{r}_{N_e}) = E\Psi, \quad (2.6)$$

where Ψ is the many-body wave-function and $\hat{H}$ is the electronic Hamiltonian,

$$\hat{H} = -\sum_{i=1}^{N_e} \frac{\hbar^2}{2m_e} \nabla_i^2 + \sum_{i=1}^{N_e} V_{\text{ext}}(\mathbf{r}_i) + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{|\mathbf{r}_i - \mathbf{r}_j|}. \quad (2.7)$$

Directly solving this equation is infeasible for systems larger than a few atoms due to the exponential scaling of the wave-function with $3N_e$ degrees of freedom.

DFT is built upon two foundational theorems established by Hohenberg and Kohn [18];

1. The ground-state electron density $n(\mathbf{r})$ uniquely determines the external potential $V_{\text{ext}}(\mathbf{r})$, and therefore all properties of the system.
2. The ground-state energy is obtained by minimising a universal energy functional $E[n]$ with respect to $n(\mathbf{r})$.

These theorems replace the many-body wave-function with the much simpler electron density, which depends only on three spatial variables.

Although the Hohenberg–Kohn theorems guarantee the existence of a universal functional, its exact form is unknown. Kohn and Sham introduced a practical approach by mapping the interacting electron system onto a fictitious system of non-interacting electrons that reproduces the same density [17].

The Kohn–Sham equations are

$$\left[-\frac{\hbar^2}{2m_e} \nabla^2 + V_{\text{eff}}(\mathbf{r}) \right] \psi_i(\mathbf{r}) = \epsilon_i \psi_i(\mathbf{r}), \quad (2.8)$$

where the effective potential is;

$$V_{\text{eff}}(\mathbf{r}) = V_{\text{ext}}(\mathbf{r}) + V_H[n](\mathbf{r}) + V_{\text{xc}}[n](\mathbf{r}). \quad (2.9)$$

Here,

V_{ext} is the external potential from the nuclei,

V_H is the Hartree (classical Coulomb) potential,

V_{xc} is the exchange–correlation (XC) potential.

The electron density is reconstructed from the Kohn–Sham orbitals;

$$n(\mathbf{r}) = \sum_{i=1}^N f_i |\psi_i(\mathbf{r})|^2. \quad (2.10)$$

where f_i is accounting for the occupation of the orbitals. The Kohn–Sham equations are solved self-consistently until the input and output densities agree within a chosen tolerance.

The exchange–correlation functional, contains all many-body effects beyond classical electrostatics. Its exact form is unknown, so practical DFT relies on approximations.

2.2.1 Plane-Wave Basis and cut-off values

The Kohn–Sham orbitals in a plane-wave basis:

$$\psi_{ik}(\mathbf{r}) = \sum_{\mathbf{G}} c_{i,\mathbf{G}} e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}, \quad (2.11)$$

where $\mathbf{G}$ are reciprocal lattice vectors as mentioned in Eq.(2.5). To cover the entire basis set requires an infinite amount of plane waves, this can not be computed without a massive cost so we have cut-off values to consider with the kinetic energy of the waves. Only plane waves with kinetic energy below a cutoff E_{cut} are included;

$$\frac{\hbar^2}{2m_e} |\mathbf{k} + \mathbf{G}|^2 \leq E_{\text{cut}}. \quad (2.12)$$

a higher energy will allow more waves into the basis set and improve accuracy but increase computational cost.

For periodic systems, integrals over the Brillouin zone are approximated using a discrete set of $\mathbf{k}$ -points. The Monkhorst–Pack scheme [19], is commonly used to generate uniform grids. Convergence with respect to $\mathbf{k}$ -point density must be verified to ensure accurate total energies and forces.

Once the electronic ground state is obtained, atomic forces are computed using the Hellmann–Feynman theorem:

$$\mathbf{F}_I = -\frac{\partial E_{\text{KS}}}{\partial \mathbf{R}_I}, \quad (2.13)$$

where $\mathbf{R}_I$ is the position of nucleus I [17]. These forces are used in *ab initio* molecular dynamics and serve as the training data for the neural network potential.

Convergence testing with respect to E_{cut} and $\mathbf{k}$ -point is therefore essential, to ensure that the base forces and energies are reliable to begin training data sets.

2.3 Neural Networks

A neural network is a parametric mathematical model designed to approximate unknown functions by composing multiple layers of linear transformations with non-linear activation functions [8].

Fig.2.1 shown here represents a standard feed-forward neural network, Formally, a feed-forward neural network consists of an input layer, one or more hidden layers, and an output layer.

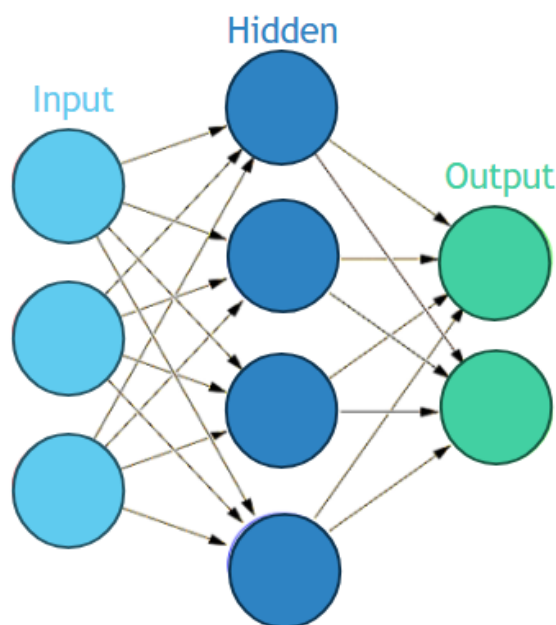


Figure 2.1: An artificial neural network (NN), showing an interconnected groups of nodes. Here, each circular node represents an artificial neuron and an arrow represents a connection from the output of one artificial neuron to the input of another.

Each layer is composed of units (or neurons) that compute weighted sums of the inputs they receive, followed by the application of a non-linear activation function.

The input layer corresponds to a set of functions that describe the local atomic environment around each atom. These inputs are passed through several hidden layers, where non-linear activation functions allow the network to approximate the PES. Finally, the output layer produces the atomic energy contribution for that atom. Summing these atomic energies yields the total energy of the system.

In atomistic modelling, the potential energy must be invariant under translation, rotation, and permutation of identical atoms. Standard Cartesian coordinates do not satisfy these invariances. Therefore, symmetry functions are used to encode the atomic environment in a way that preserves all required physical symmetries.

2.3.1 Behler-Parrinello Method

NNPs tackle this challenge by learning the PES directly from first-principles reference data. However, simply applying neural networks to atomic coordinates is fundamentally flawed because as mentioned above Cartesian coordinates do not account for the physical symmetries of the PES. The total energy of a system must remain unchanged under global translation, rotation, and permutation of identical atoms. Any representation that fails to respect these properties cannot produce a physically consistent potential.

The Behler–Parrinello (BP) framework addresses this problem by breaking down the total energy into contributions from individual atoms,

$$E_{\text{tot}} = \sum_i E_i, \quad (2.14)$$

where each atomic energy E_i depends only on the local environment of atom i within a certain cut-off radius. This approach ensures size extensive and allows for linear scaling with system size, while being general enough to represent many-body interactions rather than relying solely on pairwise additivity. Importantly, this formulation reflects the locality seen in real materials, where chemical interactions weaken with distance due to bonding effects [5].

To represent each local atomic environment while respecting physical symmetries, Behler and Parrinello created atom-centred symmetry functions. These functions map the positions of neighbouring atoms onto a fixed-length descriptor vector that respect the global invariances.

2.3.1.1 Radial symmetry functions

Radial symmetry functions encode the distribution of neighbouring atoms around a central atom as smooth functions of interatomic distances. A commonly used radial symmetry function is defined as

$$G_i^2 = \sum_{i \neq j} e^{-\eta(r_{ij}-R_s)^2} f_c(r_{ij}) \quad (2.15)$$

where r_{ij} denotes the radial distance between neighbouring atoms i and j and f_c providing a smooth cut off function.

2.3.1.2 Angular symmetry functions

Radial symmetry functions alone are insufficient to uniquely characterise local atomic environments, as distinct geometries may share identical radial distributions. Angular symmetry functions therefore incorporate three-body correlations by measuring the angles formed by triplets of atoms. One of the angular symmetry function chosen is the narrow angle given by

$$G_i^3 = 2^{1-\zeta} \sum_{i \neq j, k} (1 + \lambda \cos \theta_{ijk}) e^{-\eta((r_{ij}-R_s)^2 + (r_{ik}-R_s)^2 + (r_{jk}-R_s)^2)} f_c(r_{ij}) f_c(r_{ik}) f_c(r_{jk}) \quad (2.16)$$

where θ_{ijk} is the angle formed by atoms j , i , and k . Wide Angular Symmetry: This is a variant of the angular symmetry function. It omits the r_{jk} term, reducing the cost while still capturing essential angular information. Wide-angular functions are particularly useful for materials with extended coordination environments, such as silicate chains. It is shown as :

$$G_i^9 = 2^{1-\zeta} \sum_{i \neq j, k} (1 + \lambda \cos \theta_{ijk}) e^{-\eta((r_{ij}-R_s)^2 + (r_{ik}-R_s)^2)} f_c(r_{ij}) f_c(r_{ik}) \quad (2.17)$$

A comprehensive guide of all the terms required to compute these results are given in Appendix B.

2.3.2 Committee Neural Network Potentials

Instead of a single model, multiple models are trained independently over the same data but using different initial weights to form a committee that offers several benefits. A single neural network can over-fit or develop biases based on the specific training trajectory. By contrast, a committee of networks provides several advantages such as committee disagreement with uncertainty's and allows for there to be an error estimation. This approach follows the methodology described by Schran et al. [7], where an active learning approach is used on the committee to further improve the reliability of the model; by running calculations, finding areas with disagreement and, retraining the models in the committee to further understand how the atoms interact within configuration space.

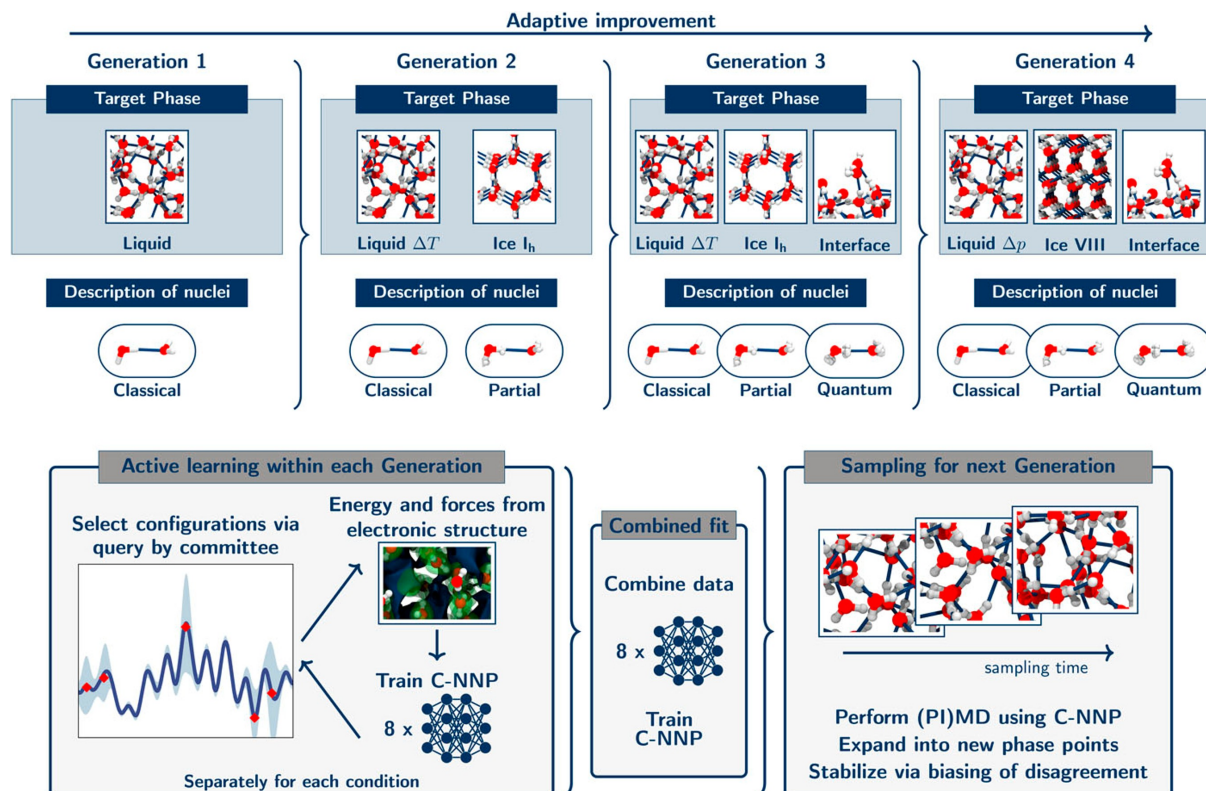


Figure 2.2: Illustration of the adaptive improvement of the committee NNP over multiple generations. The top panel summarizes the expansion into new target phases and the iterative improvement of the description of the nuclei in each generation. Within each generation, the most important points for an improvement of the model are actively selected based on the highest committee disagreement as mentioned in Section 1.2, separately for each selected state point (bottom left). Afterward, the reference energy and forces, if previously unknown for these structures, are obtained from explicit electronic structure calculations (In this study recompute DFT values). These points are added to the training set, and the committee members are trained to the expanded training set. QbC iterations are repeated until the committee disagreement reaches a given tolerance. At the end of each generation, all training points are gathered in order to perform a final extended fit of the committee model (bottom middle). The resulting C-NNPs can consecutively be applied for exhaustive (PI)MD sampling at various new state points (bottom right). These simulations provide the structures for the next generation in the adaptive improvement [7].

2.4 The δ -Learning Approach

The δ -learning strategy provides an alternative methodology for constructing machine-learning force fields by decomposing the total potential energy into a simple baseline contribution and a machine-learned correction term. Rather than training a neural network to reproduce the full DFT potential energy surface directly, the network is trained only on the difference between the DFT reference data and an analytically defined baseline potential this so called 'delta data'. This difference constitutes the δ -term from which the method takes its name.

In the present work, the Lennard–Jones (LJ) potential was selected as the baseline. The LJ form captures the essential short-range repulsive behaviour where two interacting particles repel each other at very close distances, attract each other at moderate distances, and eventually stop interacting at infinite distance. and is given by

$$V(r) = 4\epsilon \left[\left(\frac{\sigma}{r} \right)^{12} - \left(\frac{\sigma}{r} \right)^6 \right]. \quad (2.18)$$

So produces a strongly repulsive form at short interatomic separations while maintaining a smooth decay at larger distances. The cut-off radius for this potential was selected with reference to the O–H bond length, approximately $0.95\text{\AA}$, ensuring that the baseline potential remains physically meaningful in the region most relevant to vibrational motion. For each configuration in training sets, the LJ energy and forces are computed and subtracted from the corresponding DFT values to obtain the δ -targets:

$$\delta E = E_{\text{DFT}} - E_{\text{LJ}}, \quad \delta F = F_{\text{DFT}} - F_{\text{LJ}}. \quad (2.19)$$

These δ -energies and δ -forces form the supervised training labels for the neural network. Since the LJ potential already captures the dominant short-range repulsion, the δ -term is generally smoother and of smaller magnitude than the raw DFT values. Consequently, the neural network is tasked with learning only the correction necessary to bring the baseline model into agreement with DFT, rather than the entire PES. This reduces the complexity of the regression problem and often improves behaviour in regions where training data are sparse.

During inference, the final predicted energy and forces combine the LJ baseline with the neural-network output:

$$E_{\text{MLFF}} = E_{\text{LJ}} + \delta E_{\text{NN}}, \quad F_{\text{MLFF}} = F_{\text{LJ}} + \delta F_{\text{NN}}. \quad (2.20)$$

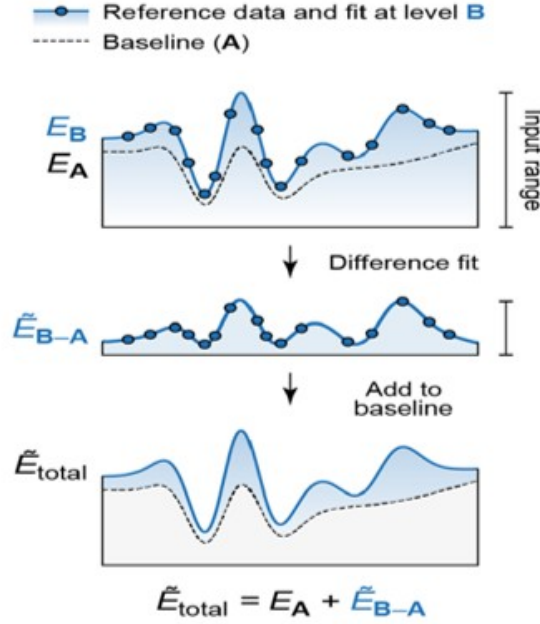


Figure 2.3: Using a lower-level baseline model, which might be a simple analytical term that only describes certain aspects of the PES (e.g., pair repulsion, fixed-charge electrostatics, or London dispersion) or a fast semiempirical method. The baseline model is subtracted from the reference data before the fit, resulting in a difference model, $\tilde{E}_{B-A}$, to which the baseline model E_A is then added back when predictions are made [9].

2.5 Thermal Properties

2.5.1 Velocity Autocorrelation Function

The velocity autocorrelation function (VACF) is a fundamental dynamical quantity in statistical mechanics and molecular dynamics. It measures the degree to which the velocity of a system at time t remains correlated with its velocity at a later time $t + \tau$, thereby characterising the atomic motion [15].

For a system of N atoms with velocities $\mathbf{v}_i(t)$, we define the full velocity vector;

$$\mathbf{v}(t) = (v_{1x}(t), v_{1y}(t), v_{1z}(t), \dots, v_{Nx}(t), v_{Ny}(t), v_{Nz}(t)) \in \mathbb{R}^{3N}, \quad i = 1, 2, \dots, N \quad (2.21)$$

The equilibrium time-correlation function then gives the VACF;

$$C(\tau) = \langle \mathbf{v}(t) \cdot \mathbf{v}(t + \tau) \rangle_t, \quad (2.22)$$

where $\langle \cdot \rangle_t$ denotes a time average over all possible time origins along the trajectory, which assuming the system is stationary and ergodic will provides an estimator of the equilibrium ensemble average of the VACF.

In a discrete molecular dynamics trajectory sampled at times; $t_k = k\Delta t$ for $k = 0, 1, \dots, T - 1$, the VACF is then estimated as;

$$C(k) = \frac{1}{3N(T-k)} \sum_{n=0}^{T-k-1} \mathbf{v}(n) \cdot \mathbf{v}(n+k), \quad (2.23)$$

where k is the lag index corresponding to a physical time $\tau = k\Delta t$. The factor $3N$ accounts for averaging over all velocity components.

It is conventional to normalise the VACF by its zero-lag value,

$$\tilde{C}(k) = \frac{C(k)}{C(0)}, \quad (2.24)$$

so that $\tilde{C}(0) = 1$. This produces a dimensionless function that facilitates comparison between different systems and simulation conditions.

The VACF provides insight into oscillations and reflect the vibrational motion, and can thus be used to calculate the Vibrational Density of States (VDOS).

2.5.2 Vibrational Density of States (VDOS)

The vibrational density of states (VDOS) describes the distribution of vibrational modes as a function of frequency. In classical molecular dynamics, the VDOS can be obtained directly from the Fourier transform of the VACF [20].

Starting from the normalised VACF in Eq.(2.24), the VDOS is defined as;

$$g(\omega) \propto \int_{-\infty}^{\infty} \tilde{C}(t) e^{i\omega t} dt. \quad (2.25)$$

As the VACF has been normalised;

$$\int_0^{\infty} g(\omega) d\omega = 3N \quad (2.26)$$

The VDOS obtained in this way is assumed not to depend on temperature and reflects the vibrational spectrum of the system: low-frequency features correspond to acoustic modes, while high-frequency peaks arise from optical modes or intramolecular vibrations. Thus, the Fourier transform of the VACF provides a direct and computationally efficient route to extracting vibrational information from molecular dynamics simulations.

2.5.3 Heat Capacity

Heat capacity is defined as the amount of heat that must be supplied to an object to produce a unit change in its temperature [21]. The VDOS $g(\omega)$, provides a direct route to computing the vibrational contribution to the heat capacity of a solid. Within the harmonic approximation, each normal mode of frequency ω behaves as an independent quantum harmonic oscillator. The internal energy associated with a single mode of frequency ω at temperature T is;

$$\varepsilon(\omega, T) = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1}, \quad (2.27)$$

where k_B is the Boltzmann constant and $\hbar$ is the reduced Planck constant [20].

For a system with vibrational density of states $g(\omega)$, normalised as shown in Eq.(2.26)

the total vibrational internal energy is

$$U(T) = \int_0^\infty g(\omega) \varepsilon(\omega, T) d\omega. \quad (2.28)$$

The vibrational heat capacity at constant volume, $C_V(T)$, is then obtained from the temperature derivative of the internal energy,

$$C_V(T) = \left(\frac{\partial U}{\partial T} \right)_V. \quad (2.29)$$

Carrying out the derivative yields the standard expression

$$C_V(T) = \int_0^\infty g(\omega) k_B \left(\frac{\hbar\omega}{k_B T} \right)^2 \frac{\exp\left(\frac{\hbar\omega}{k_B T}\right)}{\left[\exp\left(\frac{\hbar\omega}{k_B T}\right) - 1 \right]^2} d\omega. \quad (2.30)$$

Eq.(2.30) provides a direct link between the VDOS obtained from molecular dynamics (via the Fourier transform of the VACF) and the temperature-dependent heat capacity within the harmonic approximation.

2.5.4 Debye Model and Debye Temperature

The Debye model provides a simplified, yet remarkably successful, description of the vibrational properties of crystalline solids [10]. In this model, the vibrational spectrum is approximated by a continuous distribution of acoustic phonon modes with a linear dispersion relation up to a maximum cutoff frequency, the Debye frequency ω_D . The Debye temperature Θ_D is defined in terms of the Debye frequency as;

$$\Theta_D = \frac{\hbar\omega_D}{k_B} \quad (2.31)$$

Here $g_D(\omega)$ is the Debye vibrational density of states, defined as a quadratic function of frequency up to the Debye cutoff ω_D , and normalised such that $\int_0^{\omega_D} g_D(\omega) d\omega = 3N$. Its explicit form is

$$g_D(\omega) = \begin{cases} \frac{9N}{\omega_D^3} \omega^2, & 0 \leq \omega \leq \omega_D, \\ 0, & \omega > \omega_D. \end{cases}$$

Substituting $g_D(\omega)$ into Eq.(2.30) yields the Debye heat capacity,

$$C_V^D(T) = 9Nk_B \left(\frac{T}{\Theta_D} \right)^3 \int_0^{\Theta_D/T} \frac{x^4 e^x}{(e^x - 1)^2} dx, \quad (2.32)$$

where $x = \hbar\omega/(k_B T)$. At low temperatures ($T \ll \Theta_D$), this expression reduces to the well-known T^3 law [10],

$$C_V^D(T) \simeq \frac{12\pi^4}{5} Nk_B \left(\frac{T}{\Theta_D} \right)^3. \quad (2.33)$$

At high temperatures ($T \gg \Theta_D$). Using $e^x - 1 \approx x$ if $|x| \ll 1$ leads to,

$$C_V^D(T) \simeq 3Nk_b \quad (2.34)$$

which is the Dulong-Petit law [10].

2.5.4.1 Determination of the Debye Temperature

In practice, the Debye temperature Θ_D can be estimated **From low-temperature heat capacity:** By fitting the low-temperature behaviour of the computed (or experimental) heat capacity $C_V(T)$ to the Debye T^3 law,

$$C_V(T) \propto T^3, \quad (2.35)$$

In the context of molecular dynamics, a common approach is to compute the VDOS from the velocity autocorrelation function, evaluate the heat capacity via Eq.(2.30), and, if desired, fit the resulting $C_V(T)$ to the Debye model in Eq.(2.32) to obtain an effective Debye temperature for the simulated system.

Chapter 3

Methods and Development

3.1 Ab initio Calculations

To generate the data needed to train a NN on tobermorite-9Å we will have to perform DFT calculations that the PES can be learned from and subsequently the forces and energies of each individual atom. To create this we can use the software package, the Vienna Ab initio Simulation Package (VASP) . This is built upon periodic boundaries and DFT as mentioned in Eq.(2.5) and Section 2.2. Within VASP there is the ability to use pseudopotentials: When considering electrons in a system with large atoms heavier than hydrogen, and especially with $Z > 10$, it is often the case that inner electrons remain in their orbitals and do not take active role in dynamics one considers. All they do is simply to contribute to the potential in which the outer (valence) electrons move. In such case, to reduce the number of "active" electrons to be considered, one can replace the potential of the nucleus and core electrons by a pseudopotentials. The latter is chosen in such a way that the states of the active valence electrons will be the same. The gain then comes from having to consider only the valence electrons rather than all electrons in the system [22].

The computational benefits when utilizing pseudopotentials shows why the PAW pseudopotentials are used this study. As mentioned in Section 2.2.1 the first step to running DFT simulations is to optimize the set of plane waves used within VASP this basis set relies on two values : The kinetic energy of the waves and the set of k-points within the Brillouin zone [2].

3.1.1 Convergence study

Firstly, we run convergence on the plane wave cut-off value starting with a base energy of $300eV$, and we run an optimization job, increasing the value by $50eV$ until the difference, within total energy is within $1meV$ per atom. As you can see from Figure 3.1 the system starts to converges after $450eV$. From this point the change in total energy can be calculated, this can be easier seen in Table 3.1. $520eV$ is shown to be within the given tolerance, as seen in the final column. This value for the energy cutoff and its corresponding output files, are used for converging the k-points.

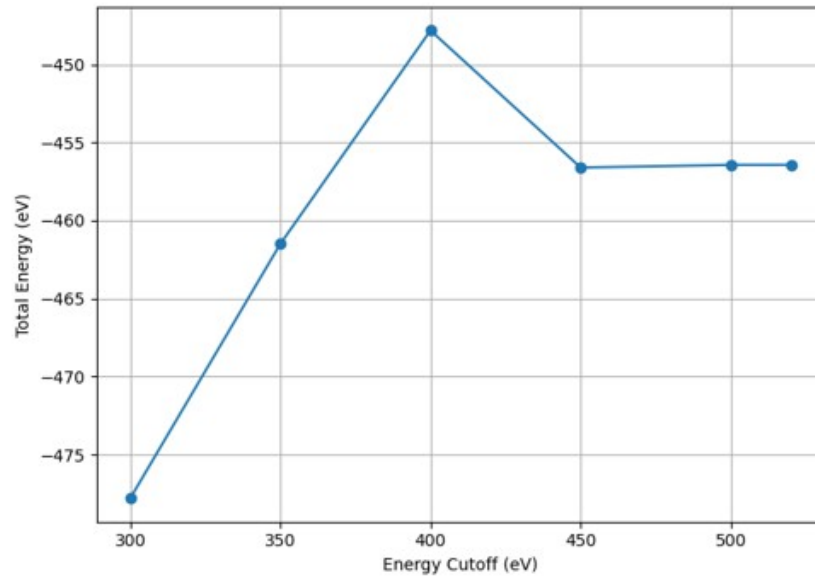


Figure 3.1: The energy cut-off against the total energy of the system; as the energy increases, we can see the system starting to converge

Energy Cut-off (eV)	Total Energy (eV)	$\Delta(eV)$
300	-477.799	
350	-461.495	16.304
400	-447.826	13.669
450	-456.610	8.784
500	-456.435	0.175
520	-456.434	0.001

Table 3.1: Total energy of the system with different values of energy cut-off and the difference between successive values

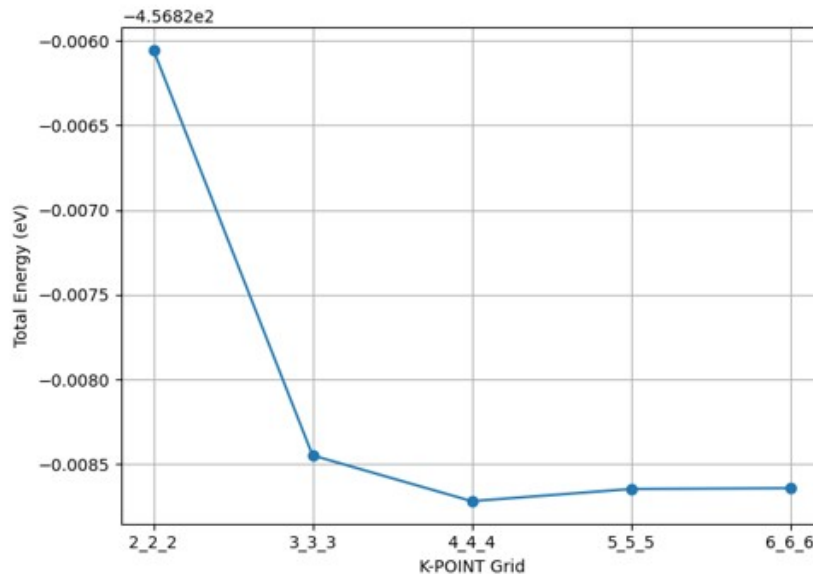


Figure 3.2: Total energy of the system with different K-point values; when we use a larger K-point value, we can see the system converging

k-points	Total Energy (eV)	$\Delta(eV)$
2_2_2	-456.8261	
3_3_3	-456.8284	2.393×10^{-3}
4_4_4	-456.8287	2.700×10^{-4}
5_5_5	-456.8286	7.174×10^{-5}
6_6_6	-456.8286	4.760×10^{-6}

Table 3.2: Total energy of the system with different values for K-points, with an energy of 520eV cut-off and the difference between successive values

Similar to the previous Convergence study the KPOINT value is increased, the difference becomes very small as the value is increased, The optimal value to select is 4.4.4 as it gives a value within the expected threshold without using too high of a computational value which would be the case using 6.6.6, an example of saving computational time without sacrificing reliability. The output files from this final study using ENCUT 520eV and KPOINTS 4.4.4 are then used as input files to run the ab initio molecular dynamics (AIMD). The code to generate these values is provided within a repository [23].

3.1.2 AIMD

Ab initio molecular dynamics (AIMD) integrates Newton's equations of motion for nuclei while forces are computed on-the-fly from DFT electronic structure shown in Section 2.2. The ground state atoms are given synthetic velocities to simulate the system at a Temperature of 300K. While running this simulation, the coordinates of each atom for steps are measured every $0.25fs$. The data used to generate the training sets require a stable temperature so the temperature is analysed as the simulation evolves.

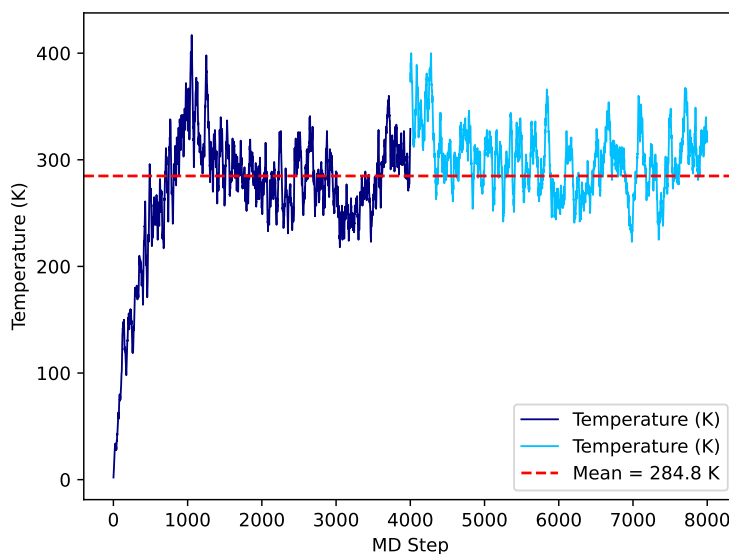


Figure 3.3: Shows how the temperature of the systems evolves using steps of 0.25fs. We can see that as the simulations continue, the system starts to be more thermally equilibrated

From Figure 3.3, as time evolves, the system becomes thermally equilibrated around the given temperature of 300K, with the AIMD being approximately 2ps in length. For the training data, the data is halved with only the last 4000 steps (1ps) being used to remove the instability of the system at the start of the AIMD.

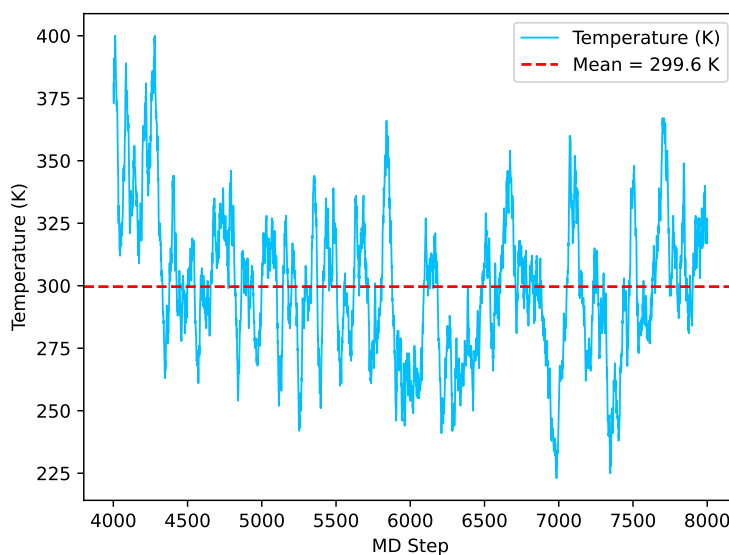


Figure 3.4: Shows the last 4000 steps of the AIMD simulations from 1000-2000fs. As seen by the mean, the system has an average of 299.6K

As seen from the graph, the mean is 299.6K, which is nearly the desired temperature of 300K from this data an Extended XYZ cartesian coordinates file can be obtained.

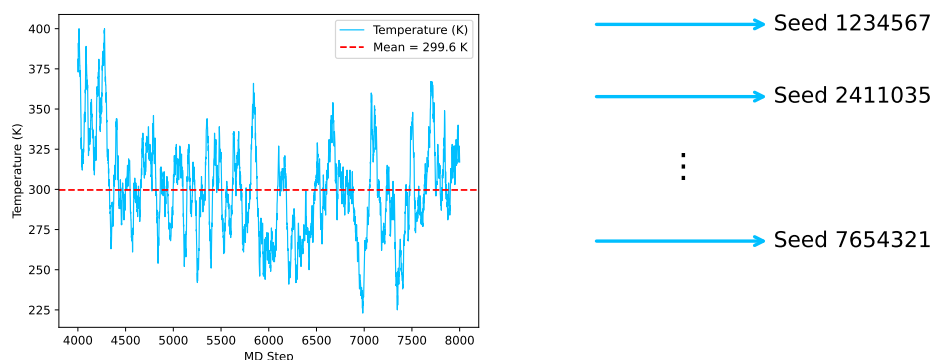


Figure 3.5: Illustration of the development of 8 independently created NN's with each seed value corresponding to the random seed variable

Called a .xyz file, that records a sequence of atomic frames; each frame lists the number of atoms followed by per-atom lines with element/type and cartesian co-ordinates ($\text{\AA}$), repeated for every time-step. An excerpt of this file is given in Appendix A. In this study the time-step is $0.25fs$ to account for the O–H bond mentioned in Section 2.1, after this data is obtained, NN models can be created.

3.2 Generating NN models

As mentioned in Section 2.3.2 this study will create a Committee NN for which the forces will be learnt from. To develop the committee multiple NN models are trained independently using different random seeds within the input files as seen in Figure 3.5. For the eight models developed, the random seed uses a pseudorandom number generator that corresponds to the number given. Each model produces a slightly different results because the initial weights, shuffling of training data, and optimisation trajectories differ due to the seed number.

Each seed will produce a slightly different model because the initial weights, shuffling of training data, and optimisation trajectories will differ as the model is trained.

With this sufficient data the models for the committee can be created. This was done through the use of the software package n2p2 (A neural network potential package) [24]. There are 3 steps required to generate the training data: scaling, pruning, rescaling and training. The workflow of this procedure will be written below but for a more comprehensive guide on creating NNs can be accessed at [25].

1. Scaling refers to transforming all symmetry-function values so that they lie within a defined, numerically stable range before they are fed into the neural network. This is necessary because raw symmetry-function values may differ by orders of magnitude, and unscaled inputs can produce unstable results. This produces a scaling.data file that is subsequently used in the next steps.
2. Pruning & Rescaling Identify and remove symmetry functions that have little effect on the training performance [25]. By pruning before training this allows for less computational time when it comes to training the model on this data. This eliminates all functions with a range (maximum - minimum value over the full inputted data, values

contained within scaling.data) below 10^{-2} . Once this has 'pruned' any redundant symmetry functions the data is needed to rescaled to ensure they remain in the numerically stable range and the training can begin.

3. Training & Retraining Neural network learns energies & forces by splitting the input data into either training or testing the model. During this training the data is placed into epochs which each contain RMSE for the model's force and energy predictions, so for each model the best epoch (lowest RMSE is selected).

Chapter 4

Results and discussion

4.1 Re-training and evolving the committee

From the investigation the reliability of the committee depends on the size, quality and diversity of the training dataset. The number of frames has to be selected to ensure that the given set of frames can capture the full configuration space. A range of frames were selected: 2000, 200 and 400. As seen in Table. 4.1, the RMSEs were calculated for both forces and energies respectively. From the table it is clear to see that 200 frames is under-fitting the model from the high RMSEs, this is due to the number of frames not being sufficiently sized to produce a reasonable model. A similar problem occurs with 2000 with the data set being too dense causing over-fitting, this is where the model is only allowing small deviations from the starting position. This causes the model to constrain the simulation and yields poor results, hence why a starting selection of 400 frames is opted to be used. At this point with a strong starting committee the active learning method discussed in Sec 2.3.2 can be utilised.

The active learning approach alone can be used to fully learn the configuration space. However, doing so would require substantial computational time and would become increasingly inefficient, each new generation yields smaller improvements while substantially expanding the dataset. So this is where the δ -learning method outlined in sec 2.4, was implemented, to speed up computational time when creating an effective committee.

#Frames	RMSE Test_Energy (<i>meV</i> per atom)	RMSE Test_Force (<i>meV</i> per Å)
2000	0.203	12.672
200	1.615	114.159
400	1.566	58.661

Table 4.1: RMSEs for both the energy and forces from the C-NNP models with a range of different amount of frames showing how number of frames can affect RMSE but ensuring that the model is not being fitted

Within the δ -learning method the inexpensive baseline chosen was the LJ potential.

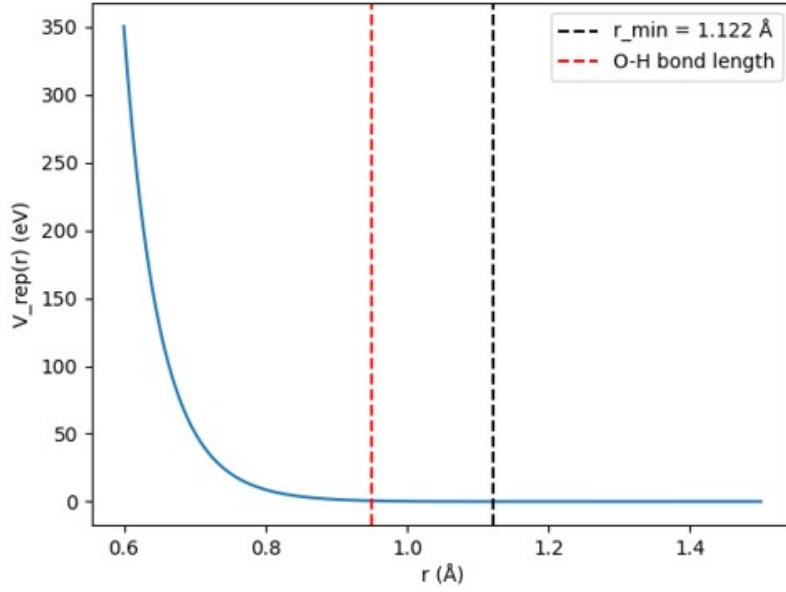


Figure 4.1: : Graph to show the LJ potential with chosen values: $\varepsilon = 0.2\text{eV}$, $\sigma = 1.0\text{\AA}$ referenced within the graph is the O–H bond found in Tobermorite-9 $\text{\AA}$

The values of sigma and epsilon chosen for Eq.(2.18) were chosen so that the LJ potential would ensure that no O–H bonds would get too small within simulations. The bonds got as small as $0.3\text{\AA}$ within early models, which is an unphysical region, so the LJ will prevent this through short range repulsion. The LJ will not affect standard O–H bonds which is shown in Fig 4.1. With any radius beyond $r_{min} = 1.122\text{\AA}$, not accounting the LJ potential, so no interactions between the Ca and Si will be affected.

For each configuration in the training set, the baseline LJ energy and forces are computed. These are subtracted from the DFT values to obtain the delta data. The neural network is then trained on these δ -values Eq.(2.19) and this new δ Committee is created. When running this new committee on LAMMPS the overlay feature is used to produce the new Energies and forces that combine both the committee and the LJ interactions shown in Eq.(2.20).

Generations	RMSE Test Energy (<i>meV</i> per atom)	RMSE Test Force (<i>meV</i> per $\text{\AA}$)
0	7.007	63.384
1	6.662	55.149
2	1.978	53.215
3	1.815	43.955
4	0.556	40.797

Table 4.2: RMSEs for both the energy and forces from the δ -learnt C-NNP models alongside the active learning approach where Generation 0 is the initial δ version

With this new approach showing longer simulations in the physical region within LAMMPS the active learning process can also be repeated to further improve this new δ learnt committee.

The importance of selecting frames that are new to the configuration space but not unphysical to undermine the real-life physics of the model is crucial to producing a robust model. A further guide to understanding the training and re-training process in successive Generations of C-NNPS can be found in Appendix C.

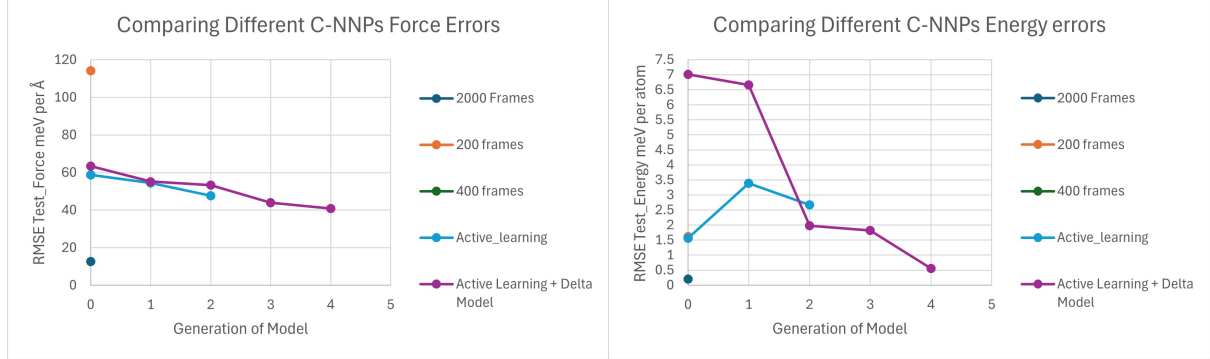


Figure 4.2: Comparison of C-NNP performance across training strategies. Left: RMSE test forces versus model generation. Right: RMSE test energies versus model generation. Frame-based models provide a good initial starting point (400 frames), while active learning shows improvement when combined with a δ -model achieving the largest and most consistent reduction in both force and energy errors.

By looking at Fig. 4.2 which encapsulates all tables used within this section and Appendix C, more easily shows the benefits of using both the active learning and δ - methods. This model allowed for long running physical simulations to be produced and from there thermal properties could be derived.

4.2 Extracting Thermal Properties

With the 4th Generational δ -model allowing long simulations within LAMMPS (Large-scale Atomic/Molecular Massively Parallel Simulator)[24], thermal properties such as the heat capacity can be calculated. As mentioned in Section 2.5, the VACF can be utilized to generate the VDOS for tobermorite-9Å.

The vibrational properties of the system were characterised using VACF, which provides a direct link between atomic dynamics in (MD) simulations and the VDOS. The VACF defined in Eq (2.22) where $\mathbf{v}(t)$ denotes the atomic velocity at time t , and the angular brackets represent an ensemble and time average. When running the simulation in LAMMPS, the velocities of each atom in cartesian co-ordinates can also be calculated, so the VACF for tobermorite 9Å can be calculated using the discrete analogue shown in Eq.(2.23).

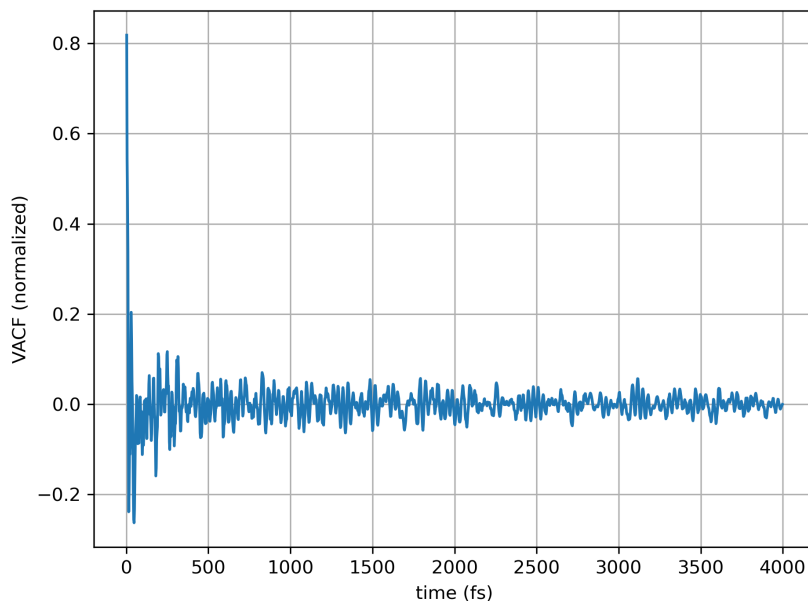


Figure 4.3: VACF of tobermorite 9Å calculated over 10ps, showing the near one initially and converging over time towards zero

The VACF is calculated over 10000fs which is 40000 frames and this allows us to have a better estimate to calculate further properties. To allow a sufficiently long LAMMPS simulation the temperature of the system is set to 50K to ensure stable trajectory integration.

4.2.1 From VACF to Vibrational Density of States

The VDOS is obtained through the Fourier transform of the VACF shown in Eq.(2.25). This transformation maps the real-time dynamical information contained in the VACF onto the frequency domain, yielding a spectrum that describes the distribution of vibrational modes in the system.

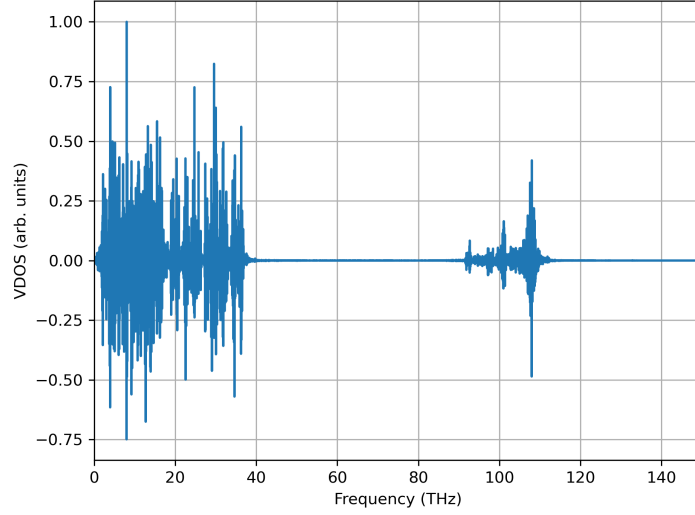


Figure 4.4: VDOS obtained from the C-NNP using the Fourier transform of VACF. The spectrum plotted as a function of vibrational frequency with significant peaks between 0 – 40 THz and around 100 – 110THz

The VDOS obtained from the VACF, can be expressed in terahertz (THz) or wave-numbers (cm^{-1}). The representation in THz is natural when working directly with simulation time and frequencies as shown in Fig. 4.4, while the conversion to wave numbers facilitates comparison with experimental vibrational spectroscopies as in Fig. 4.6. The conversion between the two frequency units is given by

$$1 \text{ THz} \approx 33.356 \text{ cm}^{-1}. \quad (4.1)$$

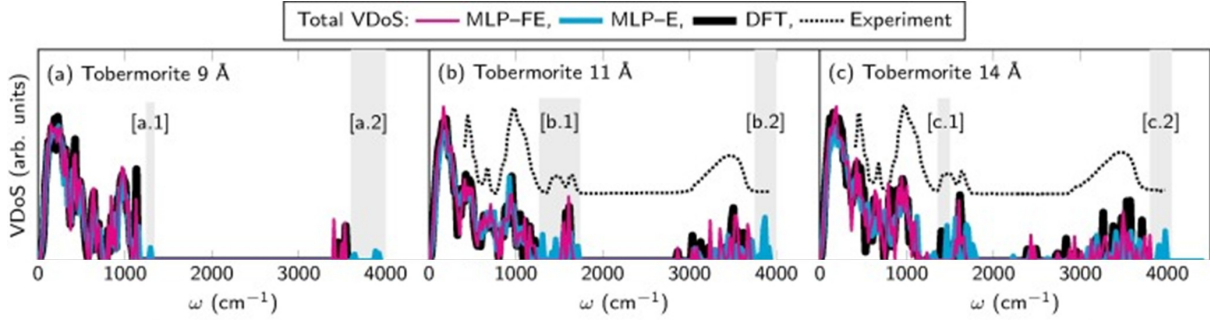


Figure 4.5: Vibrational density of states of tobermorite (a) 9 Å, (b) 11 Å, and (c) 14 Å across multiple machine learnt force fields and compared against analytically calculated values from DFT [6] .

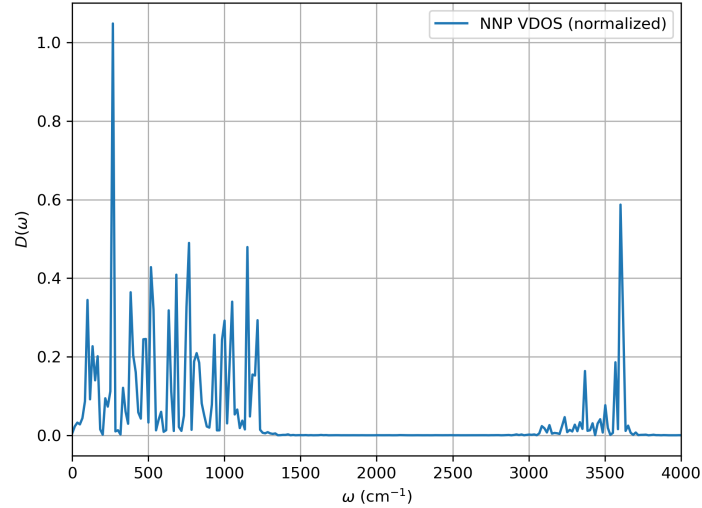


Figure 4.6: VDOS obtained from the C-NNP using the Fourier transform of VACF. The spectrum is normalised and plotted as a function of vibrational frequency with significant peaks between $0 - 1500\text{cm}^{-1}$ and around $3000 - 3600\text{cm}^{-1}$

Within Fig. 4.6 The VDOS exhibits a broad distribution of vibrational modes. Low-frequency modes correspond to collective and framework-like motions, while higher-frequency features are associated with lighter atoms and more localised vibrations. In particular, the presence of low-frequency peaks between the $0 - 1500\text{cm}^{-1}$ range is dominated by heavier atoms and network vibrations this would be caused by the bond stretches between Ca, Si and O. The high-frequency peak found around 3600cm^{-1} indicative of stretching modes involving light species so corresponds to the $O - H$ bond stretch.

By comparing the VDOS calculated in this paper to the results of Kobayashi given in Fig. 4.5 [6], the same general shape of the VDOS is given accounting for the major spikes at 1000 and 3600 cm^{-1} . This indicates that the model produced within this report is reliable and accurately modelling that of tobermorite-9Å.

4.2.2 Connection to Thermodynamic Properties

Having validated the VDOS, The VDOS can now form the basis for the calculation of the vibrational contribution to the heat capacity. Within the harmonic or quasi-harmonic approximation shown in section 2.5.3, the constant-volume heat capacity can be obtained by integrating the VDOS weighted by the appropriate quantum statistical factor as shown in Eq.(2.30).

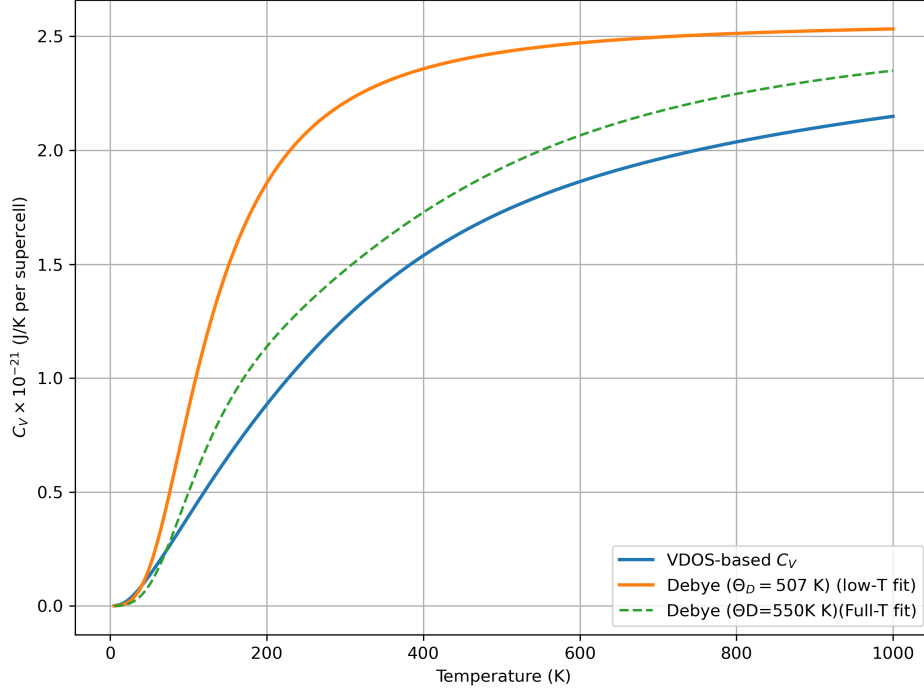


Figure 4.7: Range of heat capacities for a set of temperatures, formed from the VDOS model (blue line), the low Debye based model (orange line) or the fuller fit Debye based model (dashed green line) both of which converge as temperature increases

Fig. 4.7 compares the models of VDOS and Θ_D for low and full fitting. They match, at low temperatures $T < 50K$ the heat capacity is proportional to T^3 and at higher temperatures transitioning to Dulong-Petit law $C_v \approx 3Nk_b$. This can be validates that at $1000K$ the heat capacity is 2.150×10^{-21} for the VDOS model, 2.251×10^{-21} for the full Θ_D fit and 2.554×10^{-21} for the low Θ_D fit. All are very close to $2.568 \times 10^{-21} = 3Nk_b$ for the $N = 62$ atomic system of tobermorite-9Å. This further validates the chosen Θ_D lies between $507 - 550K$.

This (Θ_D) which was defined in Section 2.5.4. Is calculated by fitting to the VDOS for either the T^3 component of the heat capacity where $T \ll \Theta_D$ or across the section with the most accurate data, in this case the $50K - 300K$ contains the most accurate data, as this is what the C-NNP was trained on. A value for the lower Debye temperature can be extracted using Eq (2.33).

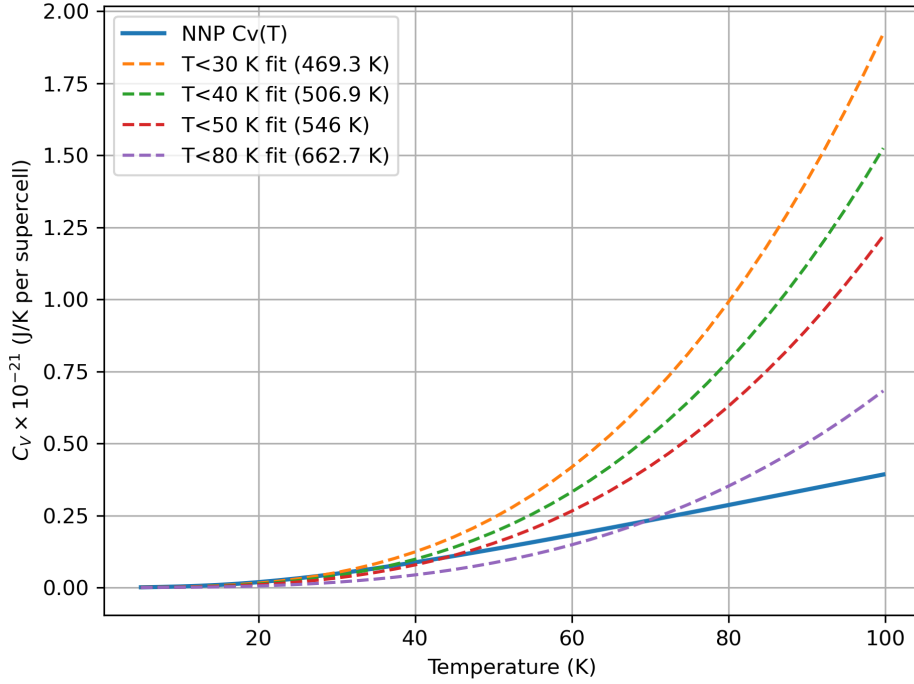


Figure 4.8: Constant-volume heat capacity of tobermorite-9Å compared with Debye T^3 fits performed over a range of different low temperatures. Along with the subsequent Debye temperatures, as the fitting window is increased the fits start to deviate from the NNP curve

By looking at Fig 4.8 the Debye temperature is estimated to be within the region of $470 - 550K$. The fit for when $T < 80K$ deviates too far from the curve to be considered. With the $T < 40K$ and $T < 50K$ fits in Fig 4.8 being the most promising fits. Comparing this to the overall fit of the data shown to be $550K$ in Fig 4.8, this model would find the $\Theta_D \approx 550K$. Although not accurate on the lower T^3 component, this matches the shape and is a more accurate representation in relation to the VDOS-based C_v .

When reviewing the literature this matches the estimated value for silicates (of which tobermorite can be considered as one) being in the region of $400 - 1000K$ [10].

This value suggested is also corroborated by Krupka [26], as the Debye temperature calculated for Wollastonite $CaSiO_3$ is $528K$. And at high temperatures $> 900K$ tobermorite-9Å starts to dehydrate and collapses into wollastonite-like phases [26] Wollastonite has a lower Debye temperature because it lacks the stiff, high-frequency local bonding environments (especially O-H and certain Si-O configurations) that raise the effective acoustic stiffness in tobermorite, further validating the choice of $550K$ for the Debye temperature.

Chapter 5

Conclusion

This report discussed the applications and benefits of creating a Machine Learning Force Field (MLFF) to model the structure and dynamics of tobermorite-9Å. By looking at the benefits of using a committee over a single NN: this allows for committee disagreement within the model to produce error estimations. Allowing the adoption of methods to speed up and improve generational models such as active learning and δ -baseline model, with the δ -learning approach, in particular, proving essential for constraining short-range interactions and accelerating the training process without sacrificing accuracy. A robust committee was created with excellent RMSEs for the energies and forces for each atom within tobermorite-9Å (0.556meV per atom and 40.797meV per Å respectively). Literature models that achieve near DFT calculation typically have RMSEs of (2meV per atom and $50\text{-}60\text{meV}$ per Å respectively [7]). Highlighting the accuracy of the Committee-Neural Network Potential used within this report. For systems of this size and complexity, performing molecular dynamics directly at the DFT level is generally impractical, as the computational cost restricts simulations to very short timescales and limited sampling. In contrast, the MLFF enables efficient force evaluations, reducing the computational cost by approximately three to five orders of magnitude [4] and thereby making large-scale and long-timescale simulations feasible.

The final committee model enabled long, stable molecular dynamics simulations from which thermal properties were extracted. The Vibrational Density Of States (VDOS) reproduced the characteristic vibrational features of tobermorite (showing peaks at 250, 400 and 3600) aligning with Ca-O, Si-O and O-H respectively. This aligns with previously published machine-learning and DFT results [6]. The heat capacity exhibited the expected transition from the Debye T^3 regime at low temperatures to the Dulong–Petit limit at high temperatures, validating the physicality of the model. From the low-temperature heat-capacity behaviour and fitting this across the full data set, a Debye temperature of approximately 550K is suggested for tobermorite-9Å. To the best of our knowledge, this value has not been explicitly reported previously in the literature for this phase. Nevertheless, this is consistent with literature values for silicate minerals and with the known thermal evolution of tobermorite toward wollastonite-like phases at elevated temperatures experimentally calculated in [26]. A further approximation for Θ_D could be calculated

by using the Einstein temperature Θ_E , which accounts for optical contributions that the Debye temperature ignores. Creating a hybrid model can capture data in the low regions of T^3 with Θ_D and capture the curvature of the C_v with Θ_E .

This work provides a theoretical value for $\Theta_D = 550K$ and establishes a robust MLFF that would allow generation of computational studies on cementitious materials, particularly those aimed at understanding and engineering their thermal behaviour. The committee-based machine-learning framework developed here can be extended to higher-hydration tobermorite phases, $11\text{\AA}$ $\text{Ca}_4\text{Si}_6\text{O}_{15}(\text{OH})_2 \cdot 5\text{H}_2\text{O}$, and $14\text{\AA}$ $\text{Ca}_5\text{Si}_6\text{O}_{16}(\text{OH})_2 \cdot 7\text{H}_2\text{O}$, which are more realistic in describing the true nature of disordered C–S–H gels. Beyond structural and vibrational properties, future studies could integrate electronic structure descriptors to explore emerging applications such as thermoelectric energy harvesting in cement composites as seen in [2], by calculating thermoelectricity from the lattice thermal conductivity. By enabling accurate prediction of vibrational spectra, heat capacity, and Debye behaviour, the present methodology provides a blueprint for evaluating theoretical modifications to cementitious materials to guide the design of next-generation sustainable concretes.

Appendices

Appendix A

Extended XYZ-co-ordinates

The creation of the extended XYZ file after running a MD file is required to be able to train a MLFF, an excerpt of the individual atoms is given below in Table. [A.2](#) The metadata (given below) accompanying the atomic coordinates specifies the simulation cell and global properties of the system. The **Lattice** field defines the three lattice vectors that describe the periodic simulation cell, while **pb**c="T T T" indicates periodic boundary conditions applied in all spatial directions. The **Properties** entry declares the per-atom data stored on each line, namely the chemical species, Cartesian positions, and atomic forces. The total potential energy and free energy of the configuration are given by **energy** and **free_energy**, respectively, and the **stress** field provides the full stress tensor of the system.

```
Lattice="11.186814 -0.009122 -0.016902  
-0.072195 7.316716 -0.007185  
-0.373759 -1.562188 9.381156"
```

```
Properties=species:S:1:pos:R:3:forces:R:3  
energy=-447.04708573  
stress="0.012733190420550168 0.0032188898109278916  
0.0018395787091989504 0.0032188898109278916  
-0.039071210494098355 -0.005532629726911067  
0.0018395787091989504-0.005532629726911067 -0.021146588684512652"  
free_energy=-447.04708573 pbc="T T T"
```

Table A.1: Atomic species, Cartesian coordinates, and forces

Atom	x (Å)	y (Å)	z (Å)	F_x (eV/Å)	F_y (eV/Å)	F_z (eV/Å)
Ca	1.37797	2.81622	0.98872	-0.129161	0.044261	0.304667
Ca	9.50241	3.04506	8.26268	-1.377822	-0.120958	0.197178
Si	0.91486	5.05791	7.37420	-0.413811	1.142632	-1.065208
O	1.58572	3.32459	3.69970	-0.019400	-0.084233	-0.951999
H	7.25040	0.62401	3.28520	-0.153307	0.052304	0.064212

Table A.2: small excerpt from the xyz file regarding the co-ordinates and forces of selected atoms within the configuration space

With the 1st column indicating which element is being looked at and the subsequent columns indicating positions and velocities in cartesian co-ordinates. The full data file for all 62 atoms across the MD can be accessed at [\[23\]](#).

Appendix B

Behler-Parrinello Symmetry functions

In the definitions of the 3 Behler–Parrinello symmetry functions shown in Eq.(2.15-2.17) The index i denotes the atom whose surroundings are being represented, while the indices j and k refer to neighbouring atoms located within the cut-off radius. The interatomic separation between atoms i and j is written as r_{ij} , and this quantity determines the point at which each neighbour contributes to the radial descriptor. The parameter η controls the width of the Gaussian function used in the radial symmetry terms; larger values of η produce narrower Gaussians and therefore allow sharper resolution of neighbours situated at specific distances. The shift parameter R_s selects the radial position at which the Gaussian is centred, and by choosing several values of R_s the symmetry functions can resolve multiple coordination shells surrounding the central atom.

For the angular symmetry functions, the angle formed at the central atom i by neighbours j and k is denoted by θ_{ijk} . This angle captures geometric information associated with three-body correlations and is particularly important in materials such as silicates where local tetrahedral or chain-like motifs play a significant structural role. The parameter λ determines the directional preference of the angular contribution. Choosing $\lambda = +1$ enhances sensitivity to configurations where the included angle is close to 0° , while selecting $\lambda = -1$ emphasises configurations where the angle is closer to 180° . The exponent ζ controls the angular resolution: increasing ζ results in a sharper dependence on deviations from the preferred angular orientation and thereby improves the descriptor’s ability to distinguish subtle differences in local geometry.

Both radial and angular symmetry functions employ a cut-off function $f_c(r)$ that smoothly reduces each neighbour’s contribution as its distance approaches the cut-off radius. The cut-off function used is;

$$f_c(r) = ((15 - 6r)r - 10)r^3 + 1 \quad (\text{B.1})$$

this ensures that the resulting energy and force predictions remain continuous and differentiable at the edge of the local environment. Seen as the differentiation of the cut-off function is $f'_c = 6x^5 - 15x^4 - 10x^3$ which for values of $r = 0, 1$ has $f'_c(0) = f'_c(1) = 0$ and $f''_c(0) = f''_c(1) = 0$ which is essential for maintaining numerical stability in molecular dynamic simulations.

Together, the quantities r_{ij} , θ_{ijk} , η , R_s , λ , and ζ determine how the symmetry functions respond to the arrangement of atoms around each atomic site. The distance and angular terms encode the geometric information, while the cut-off function enforces locality and smooth truncation. Through this construction, the symmetry functions provide a rigorous, symmetry-preserving description of the atomic environment suitable for use in Behler–Parrinello neural–network potentials.

Appendix C

Justification and discussion of model development

The choices of parameters within generating a NN is no easy task, with balancing both accuracy and computational time several factors have to be considered.

Initially creating the Committee with 2000 frames for each of the 8 NN models to learn off, as seen in Table. 4.1. Errors from this model were extremely low for both the energies and forces, indicating the model was well- behaved and could accurately reproduce the forces and energies for PES. However when using this particular committee to calculate said values within LAMMPS the simulations became unstable, and the system failed to progress for a sufficiently long time to extract meaningful thermal properties. This happened because the C-NN became too rigid a known phenomenon in neural-network potentials: Essentially the training data is dense so the model has a lot of data to work with but the data is not sufficiently diverse so over-fitting happens as the model tries to fit to the noise of the input data and hence becomes too specific that it can't generalize well to new training data.

Reducing the number of frames to 200 to avoid this over-fitting, causes the system to be under-trained instead. 200 is not a sufficiently sized data set to produce a reasonable model to simulate with LAMMPS. An effective balance between ensuring the models were not over-fitted or under-trained occurred when the selected frame count was 400.

At this point with a strong starting committee the active learning method discussed in Sec 2.3.2 can be utilised. To select the frames to retrain the model the standard error (S.E) is calculated for each individual time-step of the LAMMPS simulation

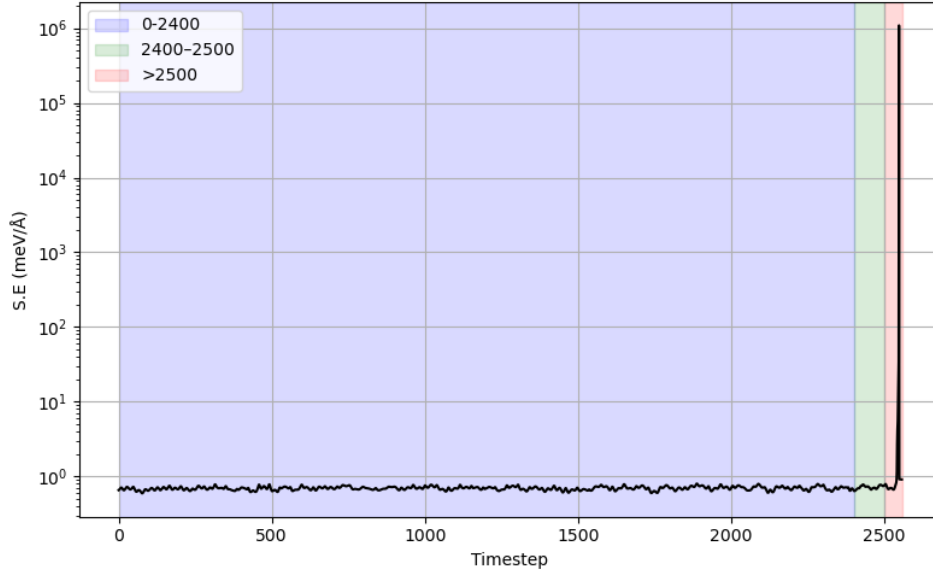


Figure C.1: shows the S.E of the model as a function of time (log scale). As the system evolves over time there is a clear indicator that the model has exploded and became unphysical, the graph is shaded into 3 regions the blue (≤ 2400) and green ($2400 - 2500$) sections are where the system is predicting the evolution of the system accurately and the final red section (≥ 2500) indicates the region of which the system becomes unphysical

The colours used in fig. C.1 are used to describe the feasibility of using the frames for re-training. The blue section (≤ 2400) has a low S.E, however, initially the model describes the configuration space that the model understands well so there is no benefit to adding them to the training set. The red section ≥ 2500 is an area of unrealistic physics so also can not be used, this leaves the green section an area where the system is entering new configuration space however not becoming unphysical.

By zooming closer towards the green section in Fig C.2 :

It is showing that around the 2550 mark is where the system will 'explode', this means a small number of frames (in the given example 40), can be selected to recompute forces and energies with DFT and be used to further train the model.

Table C.1 shows the process of using this active learning method when developing a C-NNP.

Generations	RMSE Test Energy (<i>meV</i> per atom)	RMSE Test Force (<i>meV</i> per Å)
1	1.566	58.661
2	3.394	54.537
3	2.668	47.648

Table C.1: root mean squared errors for both the energy and forces from the C-NN models showing the active learning approach with the 400-model from Table 4.1

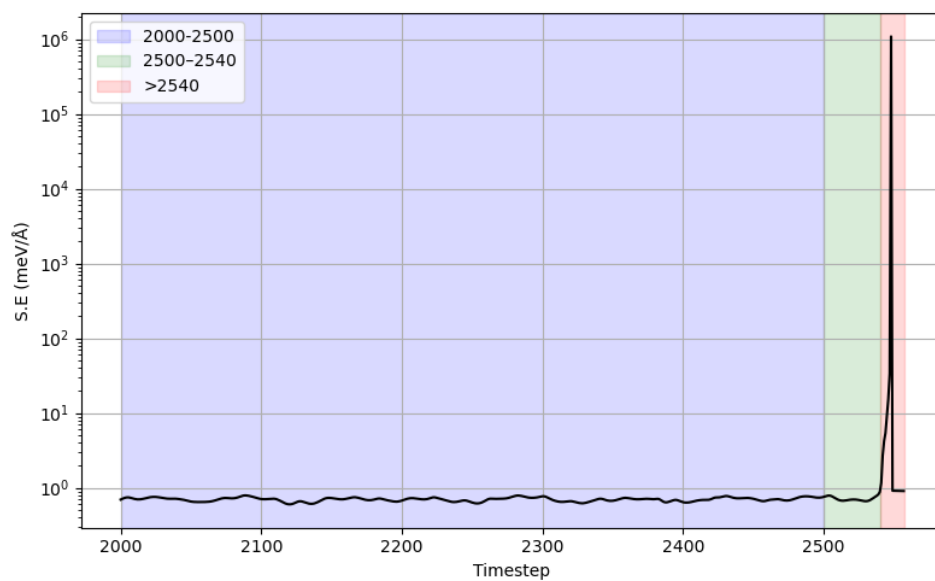


Figure C.2: shows a zoomed in version of S.E of figure C.1 The shaded regions of blue are still showing frames that are too similar to that which are already in the training set and red being ones that are unphysical. The green (2500 – 2540) sections are where the system is in new configuration space but still physically accurate

Bibliography

- [1] Luis Pérez-Lombard, José Ortiz, and Christine Pout. [A review on buildings energy consumption information](#). *Energy and buildings*, 40(3):394–398, 2008.
- [2] Esther Orisakwe, Conrad Johnston, Ruchita Jani, Xiaoli Liu, Lorenzo Stella, Jorge Kohanoff, Niall Holmes, Brian Norton, Ming Qu, Hongxi Yin, et al. [Thermoelectric properties of cement composite analogues from first principles calculations](#). *Materials Research Express*, 10(9):095505, 2023.
- [3] Ellis Gartner, Ippei Maruyama, and Jeffrey Chen. [A new model for the CSH phase formed during the hydration of Portland cements](#). *Cement and Concrete Research*, 97:95–106, 2017.
- [4] Chenglong Qiu, Tore Brinck, and Jiacheng Wang. [Modeling Potential Energy Surface by Force Fields for Heterogeneous Catalysis: Classification, Applications, and Challenges](#). *Chemical Science*, 2025.
- [5] Jörg Behler and Michele Parrinello. [Generalized neural-network representation of high-dimensional potential-energy surfaces](#). *Physical review letters*, 98(14):146401, 2007.
- [6] Keita Kobayashi, Hiroki Nakamura, Akiko Yamaguchi, Mitsuhiro Itakura, Masahiko Machida, and Masahiko Okumura. [Machine learning potentials for tobermorite minerals](#). *Computational Materials Science*, 188:110173, 2021.
- [7] Christoph Schran, Krystof Brezina, and Ondrej Marsalek. [Committee neural network potentials control generalization errors and enable active learning](#). *The Journal of Chemical Physics*, 153(10), 2020.
- [8] Wikipedia Neural network. [Neural network \(machine learning\)](#), 2026.
- [9] Volker L Deringer, Albert P Bartók, Noam Bernstein, David M Wilkins, Michele Ceriotti, and Gábor Csányi. [Gaussian process regression for materials and molecules](#). *Chemical reviews*, 121(16):10073–10141, 2021.
- [10] Richard A Robie. [Debye temperatures of selected silicate minerals](#). Technical report, US Geological Survey,, 1988.
- [11] Elena Tajuelo Rodriguez, Krassimir Garbev, Daniela Merz, Leon Black, and Ian G Richardson. [Thermal stability of CSH phases and applicability of Richardson and Groves’ and Richardson C-\(A\)-SH \(I\) models to synthetic CSH](#). *Cement and Concrete research*, 93:45–56, 2017.

- [12] Cristian Biagioni, Elena Bonaccorsi, Marco Lezzerini, and Stefano Merlino. [Thermal behaviour of Al-rich tobermorite](#). *European Journal of Mineralogy*, 28(1):23–32, 2016.
- [13] Furio Ercolessi. [A molecular dynamics primer](#). *Spring college in computational physics, ICTP, Trieste*, 19, 1997.
- [14] [State of the art in composition, fabrication, characterization, and modeling methods of cement-based thermoelectric materials for low-temperature applications](#), author=Liu, Xiaoli and Jani, Ruchita and Orisakwe, Esther and Johnston, Conrad and Chudzinski, Piotr and Qu, Ming and Norton, Brian and Holmes, Niall and Kohanoff, Jorge and Stella, Lorenzo and others. *Renewable and Sustainable Energy Reviews*, 137:110361, 2021.
- [15] Michael P Allen, Dominic J Tildesley, et al. [Computer simulation of liquids](#), 1987.
- [16] The VASP Manual. [thermostat - VASP Wiki](#), 2025.
- [17] Nathan Argaman and Guy Makov. [Density functional theory: An introduction](#). *American Journal of Physics*, 68(1):69–79, 2000.
- [18] Pierre Hohenberg and Walter Kohn. [Inhomogeneous electron gas](#). *Physical review*, 136(3B):B864, 1964.
- [19] Peter Kratzer and Jörg Neugebauer. [The basics of electronic structure theory for periodic systems](#). *Frontiers in chemistry*, 7:106, 2019.
- [20] Mohammad Javad Abdolhosseini Qomi, Franz-Josef Ulm, and Roland J-M Pellenq. [Physical origins of thermal properties of cement paste](#). *Physical Review Applied*, 3(6):064010, 2015.
- [21] David Halliday, Robert Resnick, and Jearl Walker. *Fundamentals of physics*. John Wiley & Sons, 2013.
- [22] The VASP Manual. [Pseudopotentials - VASP Wiki](#), 2025.
- [23] Niall McCallion. [A Concrete Method for Creating Neural Networks](#), 2026. GitHub Repository.
- [24] Andreas Singraber, Jorg Behler, and Christoph Dellago. [Library-based LAMMPS implementation of high-dimensional neural network potentials](#). *Journal of chemical theory and computation*, 15(3):1827–1840, 2019.
- [25] Andreas Singraber. [Welcome to the documentation of n2p2! 2p2 - A neural network potential package](#), 2016. Accessed March 2026.
- [26] Kenneth M Krupka, Richard A Robie, Bruce S Hemingway, Derrill M Kerrick, and Jun Ito. [Low-temperature heat capacities and derived thermodynamic properties of anthophyllite, diopside, enstatite, bronzite, and wollastonite](#). *American Mineralogist*, 70(3-4):249–260, 1985.